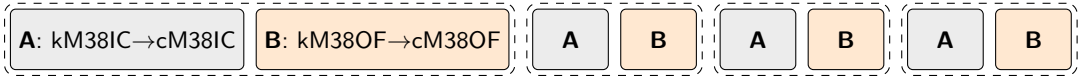
Side A: the row `ic_online_indexbuild2_r770`
source 4008a47a… (PR #1009's frozen digest, re-derived)
rank stage: `PIPE_FRAME` (point addition in the target's frame)
online scan: `PIPE_COMPACT`, batched S3 solve per candidate (about 220 ns/probe; 333 ms of 11.54 s E2E in run 39464544962)

Side B: online frame `ic_online_onlineframe2_r770`
source f07bcb1e… = 4008a47a… + 9 counted edits (invertible),
7 imported verbatim from PR #988's derivation (`KIC_ONLINE_SOLVE`)
online scan: `KIC_ONLINE_SOLVE=frame`, the rank stage's own loop
setup, index, self-test, rank stage: the parent's bytes

Same output, gated per process, per point and per base (K600, M38, M10): index and layout digests; per point one relation (indices, x codes, intermediates), one probe count, one root-call and one relation-check count across both binaries and all online solves (s3, frame, frame_s3plant); the same scalar (oracle replay); the same rank tables; per rank seed the same relation digest, probe vector, root calls and counters (kM38IC, kM38OF, kM38OFaa); online counters as the arm's (one frame; planted check only in cM38OFX).

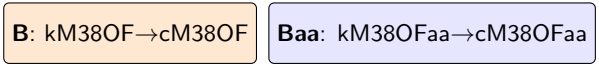
Segment Pi (point s$(i-1)$ of 16, role onlineframe2-cold-v1); sub-block order shuffled by `random.Random(2026102170+`$i-1$`)`

primary: 4 pairs, strict alternation ABAB… or BABA… (seeded start)

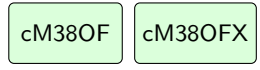


pair 1, seed $2300+4(i-1)+1$ on both sides | pair 2 | pair 3 | pair 4

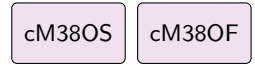
A/A: 1 pair, alternating, seed of primary pair 1



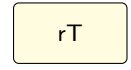
planted: frame_s3plant



layout: s3 vs frame, one binary



rho



Run order: O1 (Q0 on cPIC, cPOF, cM10IC, cM10OF, cM38IC, cM38OF, cM38OFX: both producers on all three bases; exactness only) then P1 … P16; 407 fresh processes, each pinned to one CPU of one ubuntu-24.04 runner; time guard at 165 min, job limit 180 min (any skip: invalid). Side = rank process (rank stage with the pair's seed, then the point) followed by cold process (verified table, then the point). Same rank seed on both sides of a pair, not a shared rank stage.

Primary (frozen): $E2E(\text{side}) = \text{rank_stage}(k) + \text{cold}(c)$, $R = \dfrac{\sum_{64\ \text{pairs}} E2E(A)}{\sum_{64\ \text{pairs}} E2E(B)}$; 95% two-stage paired bootstrap (points, then pairs within points; A and B together), 10,000 draws, seed 20261060. **Improvement iff the lower bound** > 1. Power: 0.91 at $R = 1.005$, about 1.00 at the modelled 1.009 (type I 0.008). Stage secondary (labelled): online cM38IC / cM38OF; layout cM38OS / cM38OF. Controls: A/A $E2E(B)/E2E(Baa)$ and cold inside $[0.97, 1.03]$; planted online cM38OFX/cM38OF, lower bound > 1; planted closure corruptions rejected. Every gate true (unknown = fail).